

Cybersecurity Incident Report

Section 1: Identify the type of attack that may have caused this network interruption

One potential explanation for the website's connection timeout error message is:
A possible cause is a **SYN flood attack**, where a malicious actor overwhelms the server with a large number of connection requests, preventing legitimate users from connecting.

The logs show that:

The server received many TCP SYN requests from multiple sources, but the connections were not fully completed. This caused a buildup of half-open connections on the server.

This event could be:

A **Denial-of-Service (DoS)** or **Distributed Denial-of-Service (DDoS)** attack targeting the web server.

Section 2: Explain how the attack is causing the website to malfunction

When website visitors try to establish a connection with the web server, a three-way handshake occurs using the TCP protocol. Explain the three steps of the handshake:

1. **SYN** – The client sends a SYN (synchronize) packet to the server requesting to start a connection.
2. **SYN-ACK** – The server responds with a SYN-ACK packet to acknowledge the request and confirm it is ready to communicate.
3. **ACK** – The client sends an ACK (acknowledgment) packet back to the server, completing the connection establishment process.

Explain what happens when a malicious actor sends a large number of SYN packets all at once:

The server attempts to respond to every SYN request with a SYN-ACK and waits for the final ACK response. If the attacker never sends the final ACK packets, the server keeps many half-open connections active. Eventually, the server's resources become overloaded, making it difficult or impossible for legitimate users to connect.

Explain what the logs indicate and how that affects the server:

The logs indicate a large volume of incomplete TCP connection requests, which is

consistent with a SYN flood attack. This overloads the server's connection queue and consumes system resources such as memory and processing power. As a result, legitimate users experience slow performance, failed connections, or website timeout errors.

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1. What do you currently understand about network attacks?

Network attacks can be grouped into a few common types:

- **Denial of Service (DoS):** Flooding a server with traffic so it can't respond to real users.
- **Distributed Denial of Service (DDoS):** Same idea, but from many different devices at once.
- **Man-in-the-middle attacks:** Intercepting communication between two parties.
- **Phishing/social engineering:** Tricking users into giving up credentials.
- **Malware-based attacks:** Infecting systems to steal or damage data.

The most relevant to your scenario is usually a traffic-flooding attack (DoS or DDoS).

2. Which type of attack would likely result in the symptoms described in the scenario?

The symptoms you described—**connection timeouts and very slow loading**—most strongly match a:

Denial of Service (DoS) or Distributed Denial of Service (DDoS) attack

More specifically, a **DDoS attack** is more likely in real-world cases because attackers usually use many machines (a botnet) to overwhelm the server.

3. What is the difference between a DoS and a DDoS?

- **DoS (Denial of Service):**
 - Comes from **one single device or source**
 - Easier to block
 - Limited attack power
- **DDoS (Distributed Denial of Service):**
 - Comes from **many devices at once (a botnet)**
 - Much harder to stop because traffic is spread out globally
 - Can generate massive traffic volume

In short:

DoS = one attacker

DDoS = many coordinated attackers

4. Why is the website taking a long time to load and reporting a connection timeout error?

This happens because the server is being overwhelmed:

- The server receives **too many requests at once**
 - It runs out of resources like:
 - bandwidth
 - CPU
 - memory
 - Legitimate user requests get stuck behind attack traffic
 - The server **cannot respond within the allowed time limit**
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1. Description of the attack (what it is + main symptoms)

The attack described is most consistent with a **Denial of Service (DoS)** or, more likely, a **Distributed Denial of Service (DDoS)** attack.

In this type of attack, the goal is not to break into the system, but to **overwhelm it with excessive traffic** so it can't respond to legitimate users.

Main symptoms/characteristics include:

- Sudden spike in network traffic (often abnormal and repetitive requests)
- Website becomes **extremely slow or completely unresponsive**
- Frequent **connection timeout errors**
- Server resources (CPU, memory, bandwidth) become maxed out
- Legitimate users are unable to access the site
- Traffic may come from many different IP addresses (especially in DDoS)

2. How the attack affected the organization's network

This attack affects the network by **flooding the web server with more requests than it can handle**.

Here's what happens step-by-step:

- Attackers send a massive number of connection requests (often SYN packets or HTTP requests)
- The server tries to process each request
- Server resources (CPU, memory, and network bandwidth) become fully consumed
- The server starts to slow down or stop responding to real users
- Valid user requests are delayed or dropped entirely

For the website specifically:

- Pages take too long to load or fail to load at all
- Sessions may disconnect unexpectedly
- The server may stop accepting new connections altogether

3. Potential consequences for the organization

A DDoS/DoS attack can cause serious operational and business damage:

Operational impacts:

- Website downtime or outages
- Disruption of online services (logins, purchases, APIs)
- IT teams must divert resources to mitigation

Financial impacts:

- Lost revenue (especially for e-commerce or service-based sites)
- Cost of emergency mitigation tools or services
- Potential SLA (service level agreement) penalties

Reputational impacts:

- Users lose trust in the reliability of the website
- Customers may switch to competitors
- Negative publicity if downtime is frequent or prolonged

Security concerns:

- Sometimes used as a distraction for other attacks (like data breaches)
- Can expose weaknesses in infrastructure

4. (Optional) How to secure the network and prevent future attacks

Organizations can reduce risk using a combination of strategies:

Traffic filtering and protection:

- Use a **DDoS protection service** (e.g., Cloudflare, AWS Shield)
- Implement **firewalls and intrusion prevention systems (IPS)**
- Block or rate-limit suspicious IP addresses

Rate limiting:

- Restrict how many requests a user/IP can send in a time period

Network architecture improvements:

- Use load balancers to distribute traffic across multiple servers
- Deploy **redundant servers** (scaling horizontally)
- Use Content Delivery Networks (CDNs) to absorb traffic spikes

Monitoring and detection:

- Monitor traffic patterns for unusual spikes
- Set up alerts for abnormal network behavior

Protocol-level protections:

- SYN flood protection (SYN cookies)

- Timeout tuning and connection limits